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Planned Programming Commands

Dataset Size and Shape:

```
print(f"Number of rows: {df.shape[0]}")
print(f"Number of columns: {df.shape[1]}")
```

Variable Types:

```
df.dtypes
df.nunique()
```

Statistical Summaries:

```
df.mean(numeric_only=True)
df.median(numeric_only=True)
df.mode().iloc[0]
df.std(numeric_only=True)
df.min(numeric_only=True)
df.max(numeric_only=True)
df.quantile([0.25, 0.50, 0.75], numeric_only=True)
df.max(numeric_only=True) - df.min(numeric_only=True)
df.quantile(0.75, numeric_only=True) - df.quantile(0.25, numeric_only=True)
df.nunique()
```

Category Frequencies:

```
df["SEX"].value_counts()
df["EDUCATION"].value_counts()
df["MARRIAGE"].value_counts()
df["PAY_0"].value_counts()
df["PAY_2"].value_counts()
df["PAY_3"].value_counts()
df["PAY_4"].value_counts()
df["PAY_5"].value_counts()
df["PAY_6"].value_counts()
```

Missingness Summary:

```
df.isna().sum()
df.isna().mean() * 100
```

Visual Analysis:

```
df["LIMIT_BAL"].plot.hist()
df["AGE"].plot.hist()

df["SEX"].value_counts().plot.bar()
df["EDUCATION"].value_counts().plot.bar()
df["MARRIAGE"].value_counts().plot.bar()

df["LIMIT_BAL"].plot.box()
df["AGE"].plot.box()

df.plot.scatter(x="LIMIT_BAL", y="AGE")

df["default payment next month"].value_counts().plot.bar()
corr = df[numerical_columns].corr()
sns.heatmap(corr, cmap="coolwarm")
```

Visual	Used to find
Histogram	Distribution of numerical data
Bar chart	Frequency of categories
Box plot	Outliers and spread
Scatter plot	Relationship between 2 numerical variables
Target distribution	Balance of target classes
Correlation heatmap	Relationships between numerical variables